Synthetic Aperture Radar Deep Statistical Imaging Through Diffusion Generative Model Conditional Inference

Zhongqi Wang, *Graduate Student Member, IEEE*, Chong Song, *Member, IEEE*, Zekun Jiao, Bingnan Wang, *Senior Member, IEEE*, and Maosheng Xiang

***Abstract*— Synthetic aperture radar (SAR) plays a crucial role in remote sensing because of its ability to operate in all weather conditions, both day and night. The traditional FFT-based SAR imaging algorithm suffers from severe speckle noise, which is almost inevitable owing to the coherent nature of the SAR system. Recently, the plug-and-play (PnP) SAR imaging method uses a plug-in denoiser as an image prior function to regularize the resulting image, thus suppressing speckle noise while maintaining the useful features of target objects. However, the existing plug-in denoisers used in statistical SAR imaging, either handcrafted or data-driven, are insufficient for complex remote sensing scenarios. More powerful image priors, such as the deep generative model for unconditional image generation, would be a better alternative regularizer for statistical SAR imaging. However, the most powerful diffusion generative model lacks an explicit latent space for conditional optimization to be adopted for SAR imaging from received signals. We propose a novel SAR imaging method based on conditional generation of a diffusion model. In detail, we embed the maximum a posteriori (MAP) formulation of SAR imaging from the received signal as a conditional guidance for diffusion generation, which overcomes the lack of latent space shortage. Compared with these statistical methods, our proposed methods exhibit exceedingly high performance both on simulated experiments and returned data imaging from RadarSat SAR data.**

***Index Terms*— Conditional generation, diffusion generative model, maximum a posteriori (MAP) estimation, statistical imaging, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) was invented for remote sensing [1]. It achieves a much higher resolution than real aperture radar, but only with a small antenna, which exceeds the diffraction limitation. In addition, it overcomes optical imaging methods because of its ability to operate in night and weather-independent functioning [2] because its working frequency is in the microwave band. Since its emergence, SAR has been widely used for remote sensing, topographical mapping, disaster/environment monitoring, and military surveillance. In recent years, SAR systems have developed a lot. One main advancement is the production of high-resolution images comparable to those of optical systems. Other paths include multiband SAR, multipolarization SAR, interferometry SAR, and tomographic SAR.

The SAR imaging principle is based on conventional radar and then generalizes it to a 2-D space. Although conventional radar can sharpen the received wave signal through matched filtering or other methods, we naturally obtain 1-D focusing. The core characteristic of SAR is the coherent processing of data collected by an antenna at locations along a trajectory in space, from which focusing along the second dimension is conducted. Hence, instead of prolonging the antenna, SAR simulates a virtual long antenna through computation.

On one hand, the coherent property is leveraged to form the synthetic aperture for better azimuth resolution as stated above. While, on the other hand, the coherent properties of SAR also cause speckle phenomena, which then appear as noises in the resulting image. The emitting microwaves have the same phase owing to the modulation. Then, after being reflected by the ground and these objects on it, highly nonlinear phases are induced by random oscillations and would later be received with the wavelength limited, which causes information loss. Thus, speckle phenomenon is intrinsically a type of chaos.

To cope with the above speckle phenomenon problems, an alternative, except for directly calculating the SAR image through a constructive imaging model, is statistical imaging. The statistical SAR imaging framework emerges from feature-enhanced SAR [3]. The author adopted the medical imaging method for SAR imaging and improved the image quality using smoothness constraints. Within the advanced digital image processing, model-based image processing tools have been developed in [4]. It uses a statistical model together with optimization-solving methods and is a powerful instrument for image processing. The plug-and-play (PnP) algorithm [5], [6], which is a typical model-based method, has several advantages. PnP methods have been applied to SAR imaging in the domain of laser SAR [7]. The author first used hand-crafted denoisers as the prior function and then used deep-learning-

Received 15 April 2024; revised 24 October 2024; accepted 11 November 2024. Date of publication 15 November 2024; date of current version 26 November 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62073306 and in part by the Youth Innovation Promotion Association, Chinese Academy of Sciences. *(Corresponding authors: Bingnan Wang; Chong Song.)*

Zhongqi Wang, Bingnan Wang, and Maosheng Xiang are with the National Key Laboratory of Microwave Imaging, Aerospace Information Research Institute, Chinese Academy of Science, Beijing 100094, China, and also with the School of Electronic, Electrical and Communication Engineering, University of Chinese Academy of Science, Beijing 100094, China (e-mail: wbn@mail.ie.ac.cn).

Chong Song and Zekun Jiao are with the National Key Laboratory of Microwave Imaging, Aerospace Information Research Institute, Chinese Academy of Science, Beijing 100094, China (e-mail: songchong@aircas.ac.cn).

Digital Object Identifier 10.1109/TGRS.2024.3498442

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:42:54 UTC from IEEE Xplore. Restrictions apply.

based denoisers to model the prior information [8]. In 2020, [9] first used the PnP framework to solve the SAR imaging problem and then obtained the best imaging results. Statistical SAR imaging [7], [9] provides a novel view of computational imaging by optimizing the corresponding maximum a posteriori (MAP) problem of the SAR imaging process. In brief, instead of the traditional sequential processing of SAR received signals, we modeled the SAR imaging process to solve a statistical problem. The goal was to maximize the conditional probability of the resulting image conditioned on the raw SAR signal. Although the likelihood term should be determined by the SAR imaging physics model, the prior term can be chosen flexibly. Hence, we can use smooth constraints as a priori to reduce speckle noise. With advanced optimization tools such as alternative direction method of multipliers (ADMM) [10], half-quadratic splitting (HQS) [11], and fast iterative soft thresholding (FIST) [12], PnP methods for SAR imaging [9] enable the substitution of the prior model with a denoiser. Deep-learning-based denoisers have brought a great leap to the statistical SAR imaging domain.

A potential breakthrough in SAR imaging is the deep generative model methods. To outline the generative model, we need to recall its counterpart in statistical learning theory, which is the discriminative model. In statistical inference, the discriminative model is meant to obtain the exact conditional distribution, which should belong to frequentists. The generative model learns the joint distribution, which originates from Bayesians. In deep learning, discriminative models for classification and segmentation are common, while generative models are less well-known. However, the generative model has a larger model capacity than the discriminative model because it models the joint distribution. Sampling from a prior distribution, according to the conditional distribution, the posterior inference can be computed. Therefore, generative inference ought to be affiliated with the framework of statistical imaging, but only with a trivial likelihood term and a much more powerful prior term. To use deep generative model in SAR imaging, we adopted the conditional generation methods of generative model inference. For generative models with a latent space, such as variational autoencoder (VAE) [13], generative adversarial network (GAN) [14], and normalizing flow (NF) [15], we can optimize in the latent space. The objective function is the distance between the generated image and target image as the condition. In [16], the authors used GAN for SAR image simulation, which was trained and inferred only on real images. Their method exceeds traditional SAR simulators by bypassing geometric assumptions and physical simplifications. However, common problems of GAN, such as mode collapse and instability, are inherited. Therefore, they adopted a clutter normalization method to stabilize the training process. The following study [17] attempted to apply a novel extension of the Wasserstein autoencoder for SAR image generation. Compared with [16], it improves the image quality and consistency. Recently, [18] designed an SAR image restoration network using the deep learning inverse technique, thus revealing the potential ability to apply deep neural networks to SAR imaging tasks. A similar study [19] also constructed a generative deep image prior network for the restoration and enhancement of SAR images. They proposed a retinex-based variational framework to enhance the low light phenomenon caused by atmospheric dispersion and reduce speckle noise.

We propose the PnP-guided conditional generation of the diffusion model for SAR imaging. The diffusion model [20], [21] is a state-of-the-art deep generative model. Our proposed method shows marvelous results in both simulated and real data experiments. To the best of our knowledge, this is the first demonstration of utilization of a diffusion generative model for SAR imaging. Previous works on applying the diffusion generative model to SAR image postprocessing are shallow implementations of the denoising diffusion probabilistic model for denoising SAR images. The pioneering work on the combination of diffusion generative model and SAR imaging was SAR-DDPM [22], which applied a diffusion model for SAR despeckling. The authors trained a noise predictor conditioned on the speckled images. During inference, they incorporated cycle spinning to improve despeckling performance. Later, [23] facilitated high-resolution image processing of SAR-DDPM by shifting and averaging patches of that image. For image-to-image translation, [24] investigated the usefulness of a conditional diffusion model for SAR-to-optical image translation. They incorporated SAR images as conditions for the training and inference processes of the diffusion model, enabling the resulting optical images to have clear borders. However, the above researches only applied diffusion generative model for simple postprocessing of SAR image despeckling, but not an integral framework of SAR imaging from received signals. The exact reason is that the diffusion generative model suffers from the big problem of no latent space for conditional generation optimization, so we cannot fully use its powerful modeling of natural images. Naturally, combining PnP statistical imaging and diffusion generative model imaging is beneficial because different views from these two paths provide the same urgent demand. For better performance, PnP statistical imaging must use a powerful prior model. However, conditional generation of the diffusion model could recur to PnP as guidance in the latent space [25], [26]. Our method deeply integrates the diffusion generative model into the SAR imaging process so that we can fully use the image modeling ability of the diffusion generative model for SAR imaging. We also show that theoretically, the diffusion-generation-model-based SAR imaging method also belongs to the framework of statistical imaging. It also outperforms the previous top algorithm, the statistical imaging methods.

This article is organized as follows. Section I is that Section II illustrate the SAR imaging geometry and its signal transformation formulation, and the main problem of speckle noise. The core algorithm combining the conditional generative diffusion model with a PnP framework for SAR imaging is presented in Section III. Finally, we present detailed experiments that compare different data and algorithms in Section IV.

II. SAR Formulation and Problems

A. SAR Observation Geometry and Signal Transformation

SAR is an active imaging device that uses electromagnetic waves in microwave bands and beyond. It emits a modulated

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:42:54 UTC from IEEE Xplore. Restrictions apply.

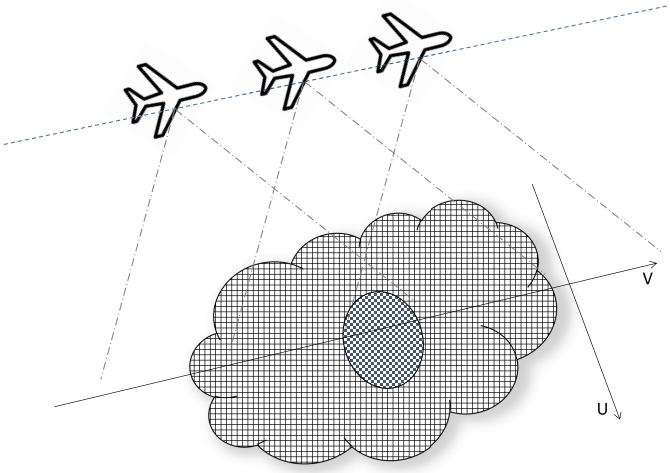


Fig. 1. Diagram of typical imaging geometry for airborne strip-map mode SAR.

electromagnetic signal based on an airborne or spaceborne platform, and then receives the echo of the emitted signal. While the signal interacts with the ground fields, the received signal contains the scattering property information of the target fields. To form an image, focusing along two dimensions must be considered as a priority, which can be accomplished by matched filtering or other reconstruction algorithms with slight differences in the directly chirp modulation along the range direction as a chirp signal according to the Doppler effect along the azimuth direction.

The imaging geometry of strip-map mode SAR [27] is shown in Fig. 1. The SAR antenna center travels along the flight path and transmits pulses perpendicular to the flight path. The most commonly used modulated signals are the linear frequency-modulated chirp signals [28]

$$s(t) = \begin{cases} e^{j(\omega_0 t + \alpha t^2)}, & |t| \le \dfrac{T_p}{2} \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

where ω_0 is the carrier frequency and 2α is the chirp rate.

For strip-map mode SAR [29], the antenna has a fixed angle and shifts along the platform. The antenna center is located at the azimuth coordinate y_0 in Fig. 1, and the returned signal of the transmitted $s(t)$ gives

$$r(y_0, t) = \int\int_{-\infty}^{\infty} f(x, y) P_L(x) P_\delta(y_0 - y) s\left(t - \frac{2R(x, y)}{c}\right) dx\ dy \tag{2}$$

where $P_L(x)P_\delta(y)$ is the assumed antenna gain pattern. We define $P_L(x) \approx 0$ for $|x| > L$, also $P_\delta(y) \approx 0$ for $|y| > \delta$.

Then, we can approximate the $R(x, y)$ term as

$$R(x, y) = \sqrt{(R + x)^2 + (y_0 - y)^2} \approx R + x + \frac{(y_0 - y)^2}{2(R + x)}. \tag{3}$$

Substituting it into the linear FM waveform $s(t)$ gives

$$\begin{aligned} r(y_0, t) \approx \mathrm{Re}\Bigg\{ & \int\int_{-\infty}^{\infty} f(x, y) P_L(x) P_\delta(y_0 - y) \\ & \cdot \exp\left[j\omega_0 \left(t - \tau_0 - \frac{2x}{c} - \frac{(y_0 - y)^2}{c(R + x)} \right) \right] \\ & \cdot \exp\left[j\alpha \left(t - \tau_0 - \frac{2x}{c} - \frac{(y_0 - y)^2}{c(R + x)} \right)^2 \right] dx\ dy \Bigg\}. \end{aligned} \tag{4}$$

The quadratic exponential term in (4) can be unfolded as

$$\begin{aligned} & \exp\left[j\alpha \left(t - \tau_0 - \frac{2x}{c} - \frac{(y_0 - y)^2}{c(R + x)} \right)^2 \right] \\ & = \exp\left[j\alpha \left(t - \tau_0 - \frac{2x}{c} \right)^2 \right] \\ & \quad \cdot \exp\left[-j\alpha \left(t - \tau_0 - \frac{2x}{c} \right) \left(\frac{(y_0 - y)^2}{c(R + x)} \right)^2 \right] \\ & \quad \cdot \exp\left[j\alpha \frac{(y_0 - y)^4}{c^2 (R + x)^2} \right] \end{aligned} \tag{5}$$

for approximately

$$-\frac{T}{2} + \frac{2L}{c} \le t - \tau_0 \le \frac{T}{2} - \frac{2L}{c}. \tag{6}$$

Therefore, the term $\|t - \tau_0 - (2x/c)\|$ is bounded by $\|(T/2)\|$. In addition, $|y_0 - y| < \delta$. Therefore, (5) would ignore the second exponential term if

$$\alpha T \frac{\delta^2}{cR} < \frac{\pi}{10} \tag{7}$$

and also ignore the third exponential term if

$$\max\left\{ \alpha \frac{(y_0 - y)^4}{c^2 (R + x)^2} \right\} = \alpha \frac{\delta^4}{c^2 R^2} < \frac{\pi}{10}. \tag{8}$$

In general, both the inequalities satisfy the bounded condition. Therefore, the return signal in (4) can be simplified to

$$\begin{aligned} r(y_0, t) \approx \mathrm{Re}\Bigg\{ & \int\int_{-\infty}^{\infty} f(x, y) P_L(x) P_\delta(y_0 - y) \\ & \cdot \exp\left\{ j\left[\omega_0 \left(t - \tau_0 - \frac{2x}{c} \right) + \alpha \left(t - \tau_0 - \frac{2x}{c} \right)^2 \right] \right\} \\ & \cdot \exp\left[-j\omega_0 \frac{(y_0 - y)^2}{c(R + x)} \right] dx\ dy \Bigg\}. \end{aligned} \tag{9}$$

So $r(y_0, (2t/c))$ has the form

$$\int\int_{-\infty}^{\infty} f(x, y) P_L(x) h(y_0 - y, t - x, x)\ dx\ dy \tag{10}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:42:54 UTC from IEEE Xplore. Restrictions apply.

where the double integral is nearly a 2-D shift-invariant convolution with a separable kernel.

Therefore, we can write the process after the FFT and stack it as phase history data

$$\begin{bmatrix} r_{y_0} \\ r_{y_1} \\ \vdots \\ r_{y_M} \end{bmatrix} = \begin{bmatrix} H_{y_0} \\ H_{y_1} \\ \vdots \\ H_{y_M} \end{bmatrix} f \tag{11}$$

where r_{y_m} is the sample vector of the received signals, H_{y_m} is the discretized matrix of the continuous observation model containing the antenna gain pattern P_L, and f is the scattering coefficient matrix of the target field.

Finally, considering that there is inevitable noise during radio propagation and discretization for approximating the continuous model, we can obtain the SAR observation model in simple linear algebra form

$$s = H \cdot f + n \tag{12}$$

where n represents the noise of all processes.

B. FFT-Based SAR Imaging Algorithms

The core idea of classical signal-processing-based SAR imaging algorithms is to focus on two dimensions, which follow the foundation of radar. Range azimuth focusing is a problem of 1-D beam sharpening which is commonly implemented by the matched filtering of modulated waves. While radar uses the Doppler effect to measure the velocity of the target object, SAR uses the Doppler effect as a source of modulated signals along the cross-range direction [1]. As a result, both range dimensional and azimuth dimensional focusing belong to the matched filtering framework [28], even in the same modulated waveform for the corresponding signal.

Although the range compression algorithm is not complex, azimuth compression has witnessed the most progress in SAR imaging algorithms. We briefly categorize this progress into error correction and complexity reduction. The former concentrates on effectiveness, whereas the latter focuses on efficiency separately. Various algorithms compensate for phase errors in the early development of SAR imaging algorithms. For example, the range–Doppler algorithm does range cell migration correction (RCMC) [30] in the range–Doppler domain to remove distortions caused by range cell migrations. The chirp scaling algorithm [31] moves forward to reduce the approximation error even more precisely by twice RCMC. The backprojection algorithm [32] ultimately takes no approximation and provides an exact phase correction factor [33] for every coordinated point. The following Ω-K [34] and parallelized backprojection algorithms [35] were proposed to satisfy the online processing demand for efficiency.

Despite these efforts for SAR imaging based on signal processing, the fundamental problem remains that these algorithms are still pseudoinverse of the observation matrix and thus have some inevitable issues, such as side production. For example, the coherent speckle phenomenon severely damages the image quality of SAR and successively confuses the following object interpretation for remote sensing.

C. Coherent Speckle Phenomenon

The speckle phenomenon is a type of fine-scale granular pattern with high fluctuations in intensity, which first appears in coherent laser systems. In addition to the coherent optical domain, it has also been observed in other coherent systems, such as SAR imagery and ultrasound medical imagery [36].

The emergence of speckles depends on the interaction of coherent electromagnetic waves after they are reflected by a sufficiently rough surface or diffused by a fuzzy diffuser ϕ, which introduces a random phase into the received signals. Based on the formulation given in (12), the SAR observations are represented in a matrix H, and f represents the preprocessed (range-compressed) signal profiles. To parsimoniously decompose the SAR observation process, we focus only on the main signal transformations

$$H = E(\phi) \cdot P \cdot T \tag{13}$$

where T is the matched filter of the SAR azimuth compression. Under the simplest mathematical formulation of inverse SAR imaging, the filter T is a DSFT transform. For more advanced linear frequency-modulated SAR imaging, T is the corresponding linear frequency modulated filter. P is the aperture or wavelength limitation of the SAR system. In the meantime, the backscattered signal comes from many elementary scatters (or facets) of the target object surface, so the distance between these scatters and the radar varies owing to the random location of the scatters. Therefore, the aperture limitation, together with the coherent property, causes the backscattered signal to interfere in the vector space(complex space) within a sampling unit (resolution cell). The received waves may be strong if wavelets are added relatively constructively, or may be weak if wavelets are out of phase. This effect causes a pixel-to-pixel variation in intensity, which manifests itself as a granular pattern called the speckle. The phase error $E(\phi)$ also contributes to intensity variation, because it mainly originates from the random diffuser during wave propagation.

Next, we need to investigate the statistics of speckle, especially the amplitude A of the complex signal $s = x\mathrm{i}+y$. Under the assumption that the number of vector sum in a resolution cell is large and random, we would call the resulted speckle as "fully developed speckle." Then, using the central limited theory, the real and imaginary components of the vector sum, x and y, are independently and identically Gaussian. Amplitude A defined as $A = (x^2 + y^2)^{1/2}$ has the Rayleigh probability distribution

$$P(A) = \frac{2A}{\sigma^2} \exp\left(-\frac{A^2}{\sigma^2}\right), \quad A \le 0. \tag{14}$$

The Rayleigh speckle model fits well with homogeneous regions in SAR images, but often fails over heterogeneous target grounds. Many other statistical distributions have also been used to model the amplitude. The Weibull and Log-normal distributions are two of the most popular examples.

Although inevitable from the viewpoint of physics and hardware design, speckles can be suppressed through software processing. Postprocessing seems to be a natural way to consider this in coherent imaging systems, but what if we

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:42:54 UTC from IEEE Xplore. Restrictions apply.

move ahead and integrate it into the imaging algorithm. Jointly suppressing speckles and imaging are the motivations for more effective imaging methods, which promotes the development of optimization-based SAR imaging methods.

Traditionally, overcoming speckles is only possible using a multilooking technique, which intrinsically averages over multiple SAR images or neighboring pixels while damaging the resolution as well. Although simple in thought, this burdens the SAR data acquisition for the demand of replicated scanning of the same region to obtain a multiimage or higher resolution than that finally acquired.

III. Diffusion-Model-Based SAR Imaging

We propose a novel SAR imaging method based on diffusion generative model conditional inference. The diffusion model is a state-of-the-art deep generative model used for unconditional image generation. We incorporate the SAR received signal as conditional guidance for diffusion generative model conditional imaging. In addition, this conditional inference can be interpreted under the PnP statistical framework, which comprises deep statistical imaging methods.

We start with a brief recall of diffusion generative model and then present the core algorithm of our proposed diffusion model conditional inference-based SAR imaging method. Finally, we relate the conditional inference imaging method to the deep statistical framework.

A. Diffusion Model

Based on annealed importance sampling (AIS) [37] and nonequilibrium thermodynamics [38], a diffusion probabilistic model [39] established a two-stage model to first diffuse the structure in a data distribution and then restore the structure by learning a reverse diffusion process.

1) Markov Chain Interpretation: Given the target data distribution $x_0 \sim q(x_0)$, the diffusion model aims to obtain the latent variable model of the form

$$p_\theta(x_0) = \int p_\theta(x_{0\cdots T})dx_{0\cdots T} \tag{15}$$

where $x_1, \ldots, x_T$ are latent variables of the same dimensionality as the data x_0. In the context of SAR imaging using a diffusion generative model, these latent variables are the intermediate variables that generate the resulting SAR image step-by-step. They are implicit and latent and do not represent explicit meaning in the SAR imaging process. Inside the integral, the joint distribution $p_\theta(X_{0\cdots T})$ is called a reverse process. For simplification, we define this process as a reversed Markov chain with a Gaussian transition kernel, starting from the stationary distribution of $p(x_T) = \mathcal{N}(x_T; 0, I)$

$$p_\theta(x_{0\cdots T}) = p(x_T)\prod_{t=1}^{T} p_\theta(x_{t-1}|x_t) \tag{16}$$

$$\text{where } p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t)\Sigma_\theta(x_t, t)). \tag{17}$$

The corresponding forward process was a settled Markov chain with a Gaussian transition kernel. Furthermore, we would give the Gaussian kernel $q(x_{1\cdots T}|x_0)$ a predefined variance schedule $\beta_1, \ldots, \beta_T$

$$q(x_{1\cdots T}|x_0) = \prod_{t=1}^{T} q(x_t|x_{t-1}) \tag{18}$$

$$\text{where } q(x_t|x_{t-1}) = \mathcal{N}\left(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t I\right). \tag{19}$$

Training approximates the variational lower bound (ELBO) of the negative log-likelihood

$$\mathbb{E}\left[-\log p_\theta(x_0)\right] \le \mathbb{E}_q\left[-\log \frac{p_\theta(x_{0\cdots T})}{q(x_{1\cdots T}|x_0)}\right] \tag{20}$$

$$= \mathbb{E}_q\left[-\log p(x_T) - \sum_{t\ge 1}\log \frac{p_\theta(x_{t-1}|x_t)}{q(x_t|x_{t-1})}\right] \tag{21}$$

$$= L. \tag{22}$$

The variance β_t can also be learned by reparameterization or predefined as a schedule. With the reparameterization notation $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t}\alpha_s$, the forward process provides each sampling x_t

$$q(x_t|x_0) = \mathcal{N}\left(x_t; \sqrt{\bar{\alpha}_t}x_0, \left(1-\bar{\alpha}_t\right)I\right). \tag{23}$$

Rewriting the loss function above would give the following objective function:

$$\mathbb{E}_q\Bigg[\underbrace{D_{KL}(q(x_T|x_0)\|p(x_T))}_{L_T}\underbrace{-\log p_\theta(x_0|x_1)}_{L_0} \tag{24}$$

$$+\sum_{t>1}\underbrace{D_{KL}(q(x_t|x_{t-1}, x_0)\|p_\theta(x_{t-1}|x_t))}_{L_{t-1}}\Bigg]. \tag{25}$$

The forward process is tractable when conditioned on x_0

$$q(x_t|x_{t-1}, x_0) = \mathcal{N}\left(x_{t-1}; \tilde{\mu}_t(x_t, x_0), \tilde{\beta}_t I\right) \tag{26}$$

$$\text{where } \tilde{\mu}_t(x_t, x_0) = \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}x_0 + \frac{\sqrt{\alpha_t}\left(1-\bar{\alpha}_{t-1}\right)}{1-\bar{\alpha}_t}x_t \tag{27}$$

$$\text{and } \tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t. \tag{28}$$

Using variational inference, we can calculate the Gaussian distribution in closed form. Using a parameterized deep neural network as a general estimator, we aim to substitute the mean and variance functions

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1};). \tag{29}$$

For the representation of the mean function $\mu_\theta(x_t, t)$, the loss function would be

$$L_{t-1} = \mathbb{E}_q\left[\frac{1}{2\sigma_t^2}\|\tilde{\mu}_t(x_t, x_0) - \mu_\theta(x_t, t)\|^2\right] + C \tag{30}$$

where C is a constant independent of θ.

The connection between the diffusion models and denoising score matching leads to a simpler stratified VAE interpretation. In detail, we reparameterize $x_t(x_0, \epsilon) = (\bar{\alpha}_t)^{1/2}x_0 + (1-\bar{\alpha}_t)^{1/2}\epsilon$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:42:54 UTC from IEEE Xplore. Restrictions apply.

for $\epsilon \sim \mathbb{N}(0, I)$ and apply the expression in (27). Then (30) would be

$$L_{t-1}$$

$$= \mathbb{E}_{x_0,\epsilon}\left[\frac{1}{2\sigma_t^2}\|\tilde{\mu}_t\left(x_t(x_0,\epsilon), \frac{1}{\sqrt{\bar{\alpha}_t}}\left(x_t(x_0,\epsilon) - \sqrt{1-\bar{\alpha}_t}\epsilon\right)\right)\right. \tag{31}$$

$$\left. - \mu_\theta(x_t(x_0,\epsilon), t)\|^2\right] + C \tag{32}$$

$$= \mathbb{E}_{x_0,\epsilon}\left[\frac{1}{2\sigma_t^2}\|\frac{1}{\sqrt{\alpha_t}}\left(x_t(x_0,\epsilon) - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon\right)\right. \tag{33}$$

$$\left. - \mu_\theta(x_t(x_0,\epsilon), t)\|^2\right] + C. \tag{34}$$

We may choose to parameterize the estimator μ_θ according to the above loss function

$$\mu_\theta(x_t, t) = \tilde{\mu}_t\left(x_t, \frac{1}{\sqrt{\bar{\alpha}_t}}\left(x_t - \sqrt{1-\bar{\alpha}_t}\epsilon_\theta(x_t)\right)\right) \tag{35}$$

$$= \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t, t)\right) \tag{36}$$

where ϵ_θ is an estimator of the residuals ϵ from x_t. Thus, the Markov transition has a closed-form expression which gives the dependence

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t, t)\right) + \sigma_t z \tag{37}$$

where $z \sim \mathcal{N}(0, I)$ is standard Gaussian noise. As a result, the only variable unknown is the gradient of data density, that is, the score function. Therefore, the training of the diffusion model resembles denoising score matching. For the loss function, there is further simplification

$$\mathbb{E}_{x_0,\epsilon}\left[\frac{\beta_t^2}{2\sigma_t^2\alpha_t(1-\bar{\alpha}_t)}\|\epsilon - \epsilon_\theta\left(\sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t\right)\|^2\right]. \tag{38}$$

This is similar to a VAE that approximates an empirical lower bound.

Indeed, denoising score matching with Langevin dynamics (SMLD) gives the same generative model, the noise conditional score network (NCSN) [40].

2) Continuous SDE Description: For a more advanced theoretical analysis of the diffusion-based generative model, we may derive a continuous description. Recall that we used a Markov chain with a Gaussian transition kernel to mathematically formulate the diffusion generative model, which can be described by the Champman–Kolmogorov equation. From this perspective, an infinitesimal time interval leads to a continuous time Markov process, which can then be described by the Kolmogorov–Feller forward/backward equation, that is, a widely used type of stochastic differential equations (SDEs).

More generally, an Itô SDE is

$$dx = f(x, t)dt + g(t)dw \tag{39}$$

where w is the Wiener process, $f(\cdot, t)$ is the drift coefficient, and $g(\cdot)$ is the diffusion coefficient of $x(t)$. The strong existence and uniqueness property of the solution of SDE states the condition of both the global Lipschitz continuity of the coefficients.

The above dedicates the forward process, and the reverse of a diffusion process is also a diffusion process, described by a reverse-time SDE

$$dx = \left[f(x, t) - g(t)^2\Delta_x \log p_t(x)\right]dt + g(t)d\bar{w}. \tag{40}$$

The additional term $\Delta_x \log p_t(x)$ is the score function of $x(t)$, and the objective function is modeled by a neural network. Note that the Wiener process $\bar{w}$ would be different from that in the forward process.

The score function can be minimized by any backbone deep neural network provided that the output is of the same dimension as the input. We can train a time-dependent score model to approximate the score function of a collection of all intermediate variables

$$\theta^* = \arg\min_\theta \mathbb{E}_t\left\{\lambda(t)\mathbb{E}_{x(0)}\mathbb{E}_{x(t)|x(0)}[\|s_\theta(x(t), t)\right. \tag{41}$$

$$\left. - \Delta_{x(t)} \log p_{0t}(x(t)|x(0))\|_2^2]\right\}. \tag{42}$$

The primary purpose of the SDE description is to obtain further experimental extensions based on theoretical analysis. The diffusion term in (40) can be simplified such that the equation becomes a deterministic process called the probability flow ODE

$$dx = \left[f(x, t) - g(t)^2\Delta_x \log p_t(x)\right]dt. \tag{43}$$

This equation enables the application of numerical solvers for differential equation, such as the Euler–Maruyama and stochastic Runge–Kutta methods. Inspired by the neural-ODE and NF, we can infer by solving (43). A black-box ODE solver can speed up the generation procedure while preserving a quality comparable to that of discrete diffusion models. In addition, deterministic latent variables make it possible to operate the latent space, such as interpolation. There have been some improvements in the diffusion model, such as the consistency model, cold diffusion model, and Poisson flow generative model. Some of them focus on the performance of the diffusion model, whereas others focus on the efficiency of the generation process. In this study, we leveraged the vanilla diffusion model. Our method can also incorporate these more advanced diffusion models.

B. Our Proposed Method of Conditional Generative Diffusion Model for SAR Imaging

We demonstrate an intuitive sketch of our proposed conditional inference diffusion generative model SAR imaging method in Fig. 2, using a pretrained unconditional diffusion generative model as the backbone of our proposed method. This method uses the powerful ability of modeling broadly

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:42:54 UTC from IEEE Xplore. Restrictions apply.

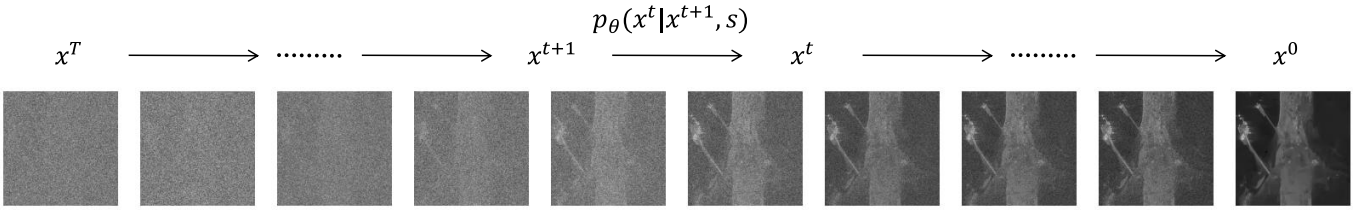


Fig. 2. Sketch of our proposed conditional inference diffusion generative model SAR imaging method. The unconditional generation inference of pure diffusion model forms the transition probability as $p_\theta(x^{t+1}|x^t)$. For SAR imaging, we use SAR received signal s to take the role of conditional guidance for the conditional inference image generation of diffusion generative model, hence the conditional probability would change to $p_\theta(x^{t+1}|x^t, s)$. As shown in the stream line, the desired SAR imaging result is generated from random distribution gradually.

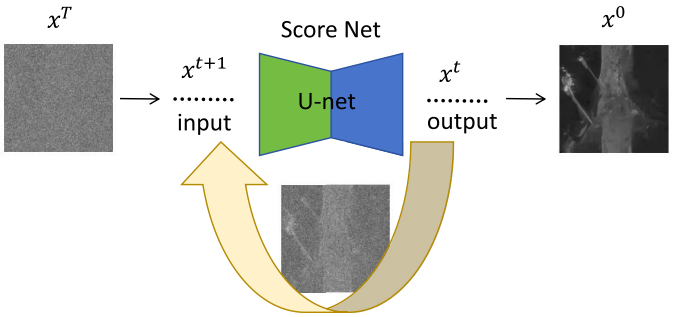


Fig. 3. Unfolded iteration process of our proposed conditional inference diffusion generative model SAR imaging method. The score net is constructed by a U-net to model the conditional distribution of x^t given x^{t+1}, i.e., the score function (log density function).

natural images of a pretrained diffusion model, instead of retraining it with a specific type of remote sensing image dataset.

For the deep learning configuration, the diffusion generative model followed the self-supervised method for deep generative training and behaved as a typical image-to-image mapping model. The diffusion-based generative model can be considered as a combination of score matching and autoregression models. First, the autoregression framework uses Langevin dynamics to construct the iteration process, as shown in Fig. 3, which resembles that of recurrent neural networks (RNNs). Second, the score matching method seeks to model the gradient of the log density distribution instead of directly modeling it. Therefore, the score net is constructed with a U-shaped net, which is widely used for image-to-image translation tasks, and it is used to model the residual of iteration, that is, $x^t - x^{t+1}$. Thus, the input and output are the intermediate variables of this autoregression model, and the output of the previous computation serves as the next input to the U-net. We used the same U-net that was used in the guided diffusion model [41].

There are two categories of conditional generation for the diffusion model: classifier-guidance and classifier-free. The classifier-guidance scheme reuses the pretrained unconditional diffusion model and newly trains a classifier for controlling the generation. On the contrary, a classifier-free scheme could be a parsimonious solution for conditional generation, which retrains the ponderous diffusion model with extra conditional control information added. Hence, an off-the-shelf classifier is much more economical for these downstream applications, compared with retraining the whole diffusion models without an extra classifier.

1) Classifier Guidance Conditional Generation: Recall that the conditional sampling process relies on the Bayesian theorem and unconditional reverse diffusion process $p_\theta(x_t|x_{t+1})$

$$p_\theta(x_t|x_{t+1}, s) = Z\, p_\theta(x_t|x_{t+1})\, p_\theta(s|x_t) \tag{44}$$

where Z is a normalizing constant and s is the received signal of an SAR system.

To sample from the above distribution, we must take the log operation and approximate $\log p_\theta(s|x_t)$ via Taylor expansion around $x_t = \mu$

$$\log p_\theta(s|x_t) \approx \log p_\theta(s|x_t)|_{x_t=\mu} \tag{45}$$

$$+ (x_t - \mu)\nabla_{x_t} \log p_\theta(s|x_t)|_{x_t=\mu} \tag{46}$$

$$= (x_t - \mu)g + C_1 \tag{47}$$

$$\text{where} \quad g = \nabla_{x_t} \log p_\theta(s|x_t)|_{x_t=\mu}, \ C_1 \text{ is a constant.} \tag{48}$$

We also take log of the transition kernel of the diffusion model

$$p_\theta(x_t|x_{t+1}) = \mathcal{N}(\mu, \Sigma) \tag{49}$$

$$\log p_\theta(x_t|x_{t+1}) = -\frac{1}{2}(x_t - \mu)^T \Sigma^{-1}(x_t - \mu) + C_2. \tag{50}$$

Incorporating the above two equations gives that

$$\log(p_\theta(x_t|x_{t+1})\, p_\theta(s|x_t)) \tag{51}$$

$$\approx -\frac{1}{2}(x_t - \mu)^T \Sigma^{-1}(x_t - \mu) + (x_t - \mu)g + C_3 \tag{52}$$

$$= -\frac{1}{2}(x_t - \mu - \Sigma g)^T \Sigma^{-1}(x_t - \mu - \Sigma g) + \frac{1}{2} g^T \Sigma g + C_4 \tag{53}$$

$$= -\frac{1}{2}(x_t - \mu - \Sigma g)^T \Sigma^{-1}(x_t - \mu - \Sigma g) + C_5 \tag{54}$$

$$= \log p(z) + C_4, \quad z \sim \mathcal{N}(\mu + \Sigma g, \Sigma). \tag{55}$$

Intuitively, the Markov transition kernel only conditions on the mean term. The extra shift intensity is determined by both the score function $\nabla_{x_t} \log p_\theta(s|x_t)$ and the variance Σ of unconditional transition kernel $p_\theta(x_t|x_{t+1})$.

Consequently, the sampling process should follow the equation:

$$\boldsymbol{x}_{t-1} = \boldsymbol{\mu}(\boldsymbol{x}_t) + \underbrace{\sigma_t^2 \nabla_{\boldsymbol{x}_t} \log p(\boldsymbol{s}|\boldsymbol{x}_t)}_{\text{addtional term}} + \sigma_t \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}). \tag{56}$$

For SAR imaging, observation geometry is an adaptable part of the imaging physics model. Therefore, the conditional module supposed that a classifier-free conditional diffusion model

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:42:54 UTC from IEEE Xplore. Restrictions apply.

would implicitly incorporate and not be settled. Although an extra classifier is tolerable, a simple change would cause the classifier to be retrained.

2) PnP Guidance Conditional Generation: Recently, [25], [26], [42] proposed the implementation of conditional generation under a PnP framework. By *plug-and-play*, we mean that the framework is capable of generating images at test time based on a replaceable prior constraint and the condition given by measurements in image restoration tasks.

The diffusion model is considered to be the prior function of the PnP framework in [25], which attempted PnP on diffusion models for various tasks by directly optimizing latent images with the off-the-shelf model. To the best of our knowledge, this article serves as the pioneering work of combining the conditional DDPM and PnP framework. However, the method in [25] fails to generate meaningful images in complex datasets such as ImageNet.

In contrast to [25], 2023CVPRPnPdiffusion developed a multiexpert strategy for conditional DDPM and successfully guided them in complex datasets by introducing small parameters into the off-the-shelf model and making it suitable for the latent.

Denoising diffusion restoration model (DDRM) [43] solves the inverse image problem by factorizing the measurement matrix through SVD. Because the image is generated from a pretrained DDPM, it can be viewed as an implicit PnP framework. The following study [26] used the differentiable loss function as guidance for conditional diffusion generation. The loss guidance provides the rectification term of gradients, whereas posterior sampling is used for conditional generation. In addition, [26] used an additional Monte Carlo approach to leverage efficiency and accuracy.

3) SAR Imaging as Condition: As mentioned above, SAR imaging can be viewed from their angles, sequentially. SAR imaging involves the flow of signal processes. Naturally, SAR is a system for active imaging; therefore, the signal can be modulated elaborately. Theoretically, every step should be designed instead of the approximation. Second, SAR imaging is a standard inverse problem, so MAP statistical modeling would be another interpretation of computational imaging. The solution method relies on optimization techniques. PnP SAR imaging also belongs to statistical modeling, with the likelihood term being trivial, and the prior term being a powerful one-step generation. Third, from the perspective of conditional generative imaging, there is a similar optimization framework for latent variables to find the closest data point among dataset distributions. The imaging results would satisfy both the measurement condition and data prior distribution condition. Hence, it would be not only qualified for reality but also vivid for visualization.

Thus, we applied the conditional generation function of the diffusion model to SAR imaging. First, the simplified imaging matrix for SAR systems can be expressed as H

$$s = H \cdot f. \tag{57}$$

It should contain all these signal processing steps, including both range and azimuth compression, as well as the necessary compensation for the curvature and residuals caused by the movement.

The classical generative interpretation of the inverse problem-solving framework can still be seen as an optimization problem, with only an implicit constraint on the support set. The basic framework still falls into the inverse problem-solving category, and the explanation of error minimization continues, but its supportive set shrinks to a specific manifold within the entire data space. Every data point on this manifold satisfies the specific properties of the corresponding image. This process can also be considered as a convex projection from the latent space to the required image space

$$\arg\min_{z} \|s - H \cdot g(z)\| \tag{58}$$

$$\hat{f} = g(\hat{z}). \tag{59}$$

Theoretically, the generated images should no longer contain speckle noise. The images appear more like optical images, which are usually captured by a camera. In addition, the generation potential helps recover these contaminated details in the acquired remote sensing images. However, we would again state that because there is no explicit latent space for the diffusion model, the trivial optimization-based conditional generation methods would fail when using the diffusion model for SAR imaging.

Recall that we derived the conditional generative diffusion function under continuous SDE interpretation

$$\boldsymbol{x}_{t-1} = \boldsymbol{\mu}(\boldsymbol{x}_t) + \underbrace{\sigma_t^2 \nabla_{\boldsymbol{x}_t} \log p(\boldsymbol{s}|\boldsymbol{x}_t)}_{\text{addtional term}} + \sigma_t \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}). \tag{60}$$

Focusing on the additional term, we can note that this can be analogous to the gradient adjustment of diffusion iteration. Mathematically, this coincides with the formulation of the likelihood function in MAP statistical modeling of the SAR imaging process. Specifically, we would consider the SAR imaging problem as a Bayesian model

$$P(\boldsymbol{x}_t|s) = \frac{P(s|\boldsymbol{x}_t)P(\boldsymbol{x}_t)}{P(s)}. \tag{61}$$

The likelihood $P(s|\boldsymbol{x}_t)$ is the distribution of the received signal conditioned on the resulting image; therefore, the likelihood models the physics model of backscattering. Then, by adopting the same Gaussian error distribution assumption for simplification, we can transform the gradient term into

$$\sigma_t^2 \nabla_{\boldsymbol{x}_t} \log p(\boldsymbol{s}|\boldsymbol{x}_t) \tag{62}$$

$$= \sigma_t^2 \nabla_{\boldsymbol{x}_t} \left[(s - H \cdot x)^2 - \sigma_\epsilon^2 - \frac{1}{2} \log(2\sigma_\epsilon) \right] \tag{63}$$

$$= -2\sigma_t^2 H^T (s - H \cdot x). \tag{64}$$

C. Diffusion Model as a Prior Term Under Statistical PnP Framework

In addition to the above interpretation, we can reconstruct the target field of reflectivity coefficients f given the returned

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:42:54 UTC from IEEE Xplore. Restrictions apply.

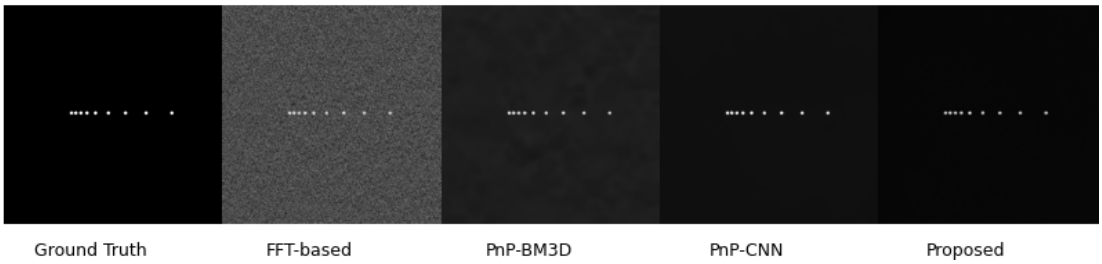


Fig. 4. Simulated experiment on point targets. We simulate several points as the target, and they distribute from concentrated to sparse, horizontally. This arrangement is used to give an accurate evaluation of the range resolution of the imaging performance of different algorithms.

echo s under the statistical PnP framework. This formulation belongs to the standard setting of inverse image problem

$$\hat{f} = \arg\min_{f} \|s - H \cdot f\|. \tag{65}$$

An extra constraint directly on the field f is also critical for the reconstruction quality of the final SAR image. This extra constraint can be naturally extended from the original inverse problem as the prior model in the corresponding MAP statistical formulation

$$\hat{f} = \arg\min_{f} \|s - H \cdot f\| + \gamma g(f). \tag{66}$$

The error minimization term is the likelihood model. Another explanation is included under an optimization framework because the inverse problem definition is exactly a minimization problem of the error in the echo space. The addition of an extra constraint converts this unconstrained optimization problem into a constrained optimization problem. An efficient solution to the constrained optimization problem would be the variable splitting methods, and the most prominent one would be the ADMM. Then, PnP further enabled the separation of the choice of prior model, so that these state-of-the-art powerful denoisers can be used to form the prior model.

Consequently, we can use the conditional generative diffusion model as the prior term $g(f)$. In detail, as we would split the optimization object through a Lagrangian multiplier, the step that contains the prior term would be a mapping from the intermediate variable space to the desired natural image space as the diffusion generative model functions as a priori.

In summary, the resulting image was optimized according to an objective function consisting of two terms. The first term is the imaging model with the required parameters of its physical settings, and this is the likelihood term in the Bayesian model. The second term is the image generative model, which serves as the prior function (or regularization function). With optimization, we want to obtain an image that not only maximizes the likelihood of the received signal through the imaging physics model but also satisfies that it follows the modeling output space of the image generative model that was pretrained with a large number of noiseless images.

IV. Experiments and Discussion

We demonstrate the performance of our proposed conditional generative diffusion model for SAR imaging on both the simulated imaging task and the real returned SAR data of RadarSat.

To the best of our knowledge, the usage of a generative model for SAR imaging has rarely appeared. However, as stated above, generative imaging may be considered a specific incorporation of the generative prior model into a PnP framework. Therefore, we compared our proposed conditional diffusion generative model for SAR imaging with traditional sequential processing methods and PnP methods using different prior models such as BM3D [44] and bias-free CNN denoisers [45]. BM3D is a hand-crafted denoiser without learning and approaches the Cramer–Rao lower bound (CRLB) for Gaussian denoising tasks [46]. For comparison, we used the BM3D library of Python implementation.[1] As for the bias-free CNN denoiser, we used official implementation[2] by the authors. The diffusion model implemented relies on the official implementation of guided-diffusion[3] by OpenAI.

A. Simulated Returned Signals

To support the superior performance of our proposed diffusion-based SAR imaging algorithm, we first conducted a simulated experiment on point targets. As shown in Fig. 4 and Fig. 5, we simulated a target field using several points. These points are distributed from concentrated to sparse to evaluate the resolution of the different algorithms. To fully assess the resolution of SAR imaging, we should experiment in two perpendicular directions: the range and azimuth directions. Here we set them in the horizontal and vertical directions. This arrangement was used to provide an accurate evaluation of the azimuth and range resolution of the imaging performance of different algorithms.

1) Visually: In Figs. 4 and 5, we show the results of different SAR imaging algorithms on horizontal and vertical resolution benchmarks. The classical FFT-based SAR imaging method causes speckle noise, which severely affects the distinction between two neighboring point targets, thus reducing the resolution of the imaging results. The other three advanced algorithms suppressed speckle noise, but their performances differed. The PnP-BM3D algorithm uses a hand-crafted BM3D denoiser as the generalized prior model, and it lacks flexibility for nonuniform noise. Hence, PnP-BM3D performs slightly outsatisfactorily, leaving too much noise. Visually, PnP-CNN

[1] https://pypi.org/project/bm3d/
[2] https://github.com/LabForComputationalVision/bias_free_denoising
[3] https://github.com/openai/guided-diffusion

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:42:54 UTC from IEEE Xplore. Restrictions apply.

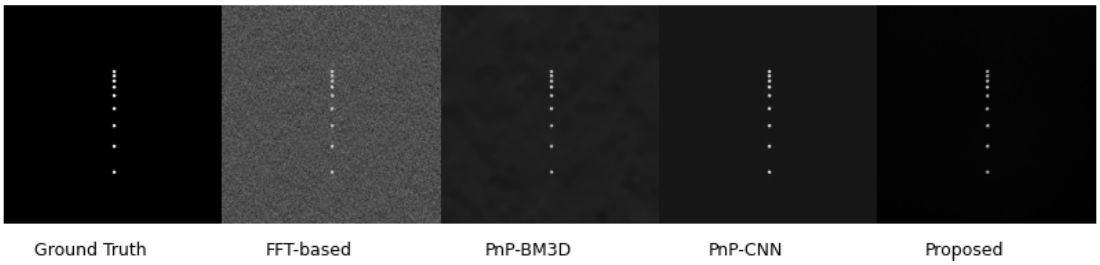


Fig. 5. Simulated experiment on point targets of vertical arrangement. We simulate several points as the target, and they distribute from concentrated to sparse, vertically. This arrangement is used to give an accurate evaluation of the azimuth resolution of the imaging performance of different algorithms.

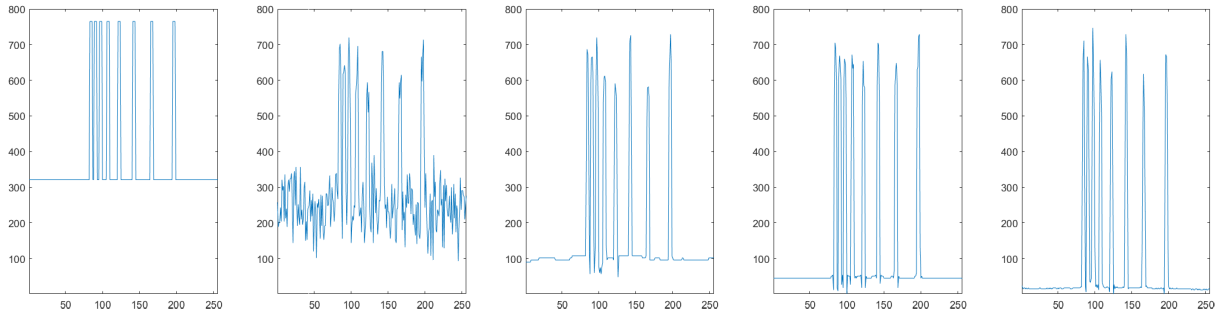


Fig. 6. Sectional curve of horizontal simulated point targets. Left to right: original point targets, classical imaging result with speckle noise, PnP-BM3D imaging result, PnP-CNN imaging result, and our proposed conditional diffusion imaging result. We plot the sectional area of horizontal simulated point targets as curves.

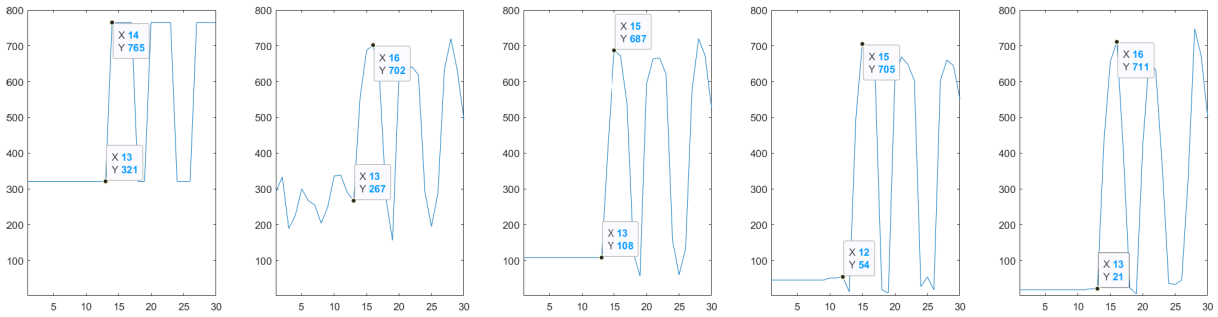


Fig. 7. Sectional curve of horizontal simulated point targets, with indicators. Left to right: original point targets, classical imaging result with speckle noise, PnP-BM3D imaging result, PnP-CNN imaging result, and our proposed conditional diffusion imaging result. We plot the sectional area of the first three horizontal simulated point targets.

and our proposed conditional diffusion both perform well, and there is no evident noise in the resulting SAR images.

2) Numerically: To evaluate the results more precisely, we need to give a quantitative assessment except for the visual quality. As shown in Fig. 6, the original image exhibits a sharp edge and flat crest for these point targets. In the speckle noised image obtained by classical FFT-based algorithm, the signal is contaminated by the noise signal, and it is difficult to distinguish two neighboring point targets. The despeckling task for this simple image of only several points is rather easy and these three advanced SAR imaging methods can distinguish different points from each other and the background. However, their suppression of background speckle noise was different. Our proposed conditional diffusion method performs the best because it acquires a cleaner background. More specifically, as shown in Fig. 7, the original image exhibited a sharp edge and flat crest of these point targets. In the speckle noised image by the classical FFT-based algorithm, we can see that not only the zenith of the point target is reduced but also the background is noised by the speckle phenomenon. These three intelligent SAR imaging methods significantly improve upon classical FFT-based methods. As a result, as the order of PnP-BM3D, PnP-CNN, and our proposed conditional diffusion, the summit of the point, as indicated in the pictures, grows taller, while the foot of the point becomes lower. Therefore, more advanced algorithm can not only reconstruct a stronger signal but also suppress more intense speckle noise.

For vertically arranged point targets, the sectional curves are shown in Figs. 8 and 9, which generally support the performance gain of our proposed conditional-diffusion-based SAR imaging method over other SAR imaging methods. The evidence is that the background of conditional diffusion imaging results is cleaner than that of other imaging results, thus making the target more extinguishable.

According to the plot curve, the relative intensity is shown in Table I.

The signal relative intensity results show that our proposed diffusion-based SAR imaging algorithm outperforms others

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:42:54 UTC from IEEE Xplore. Restrictions apply.

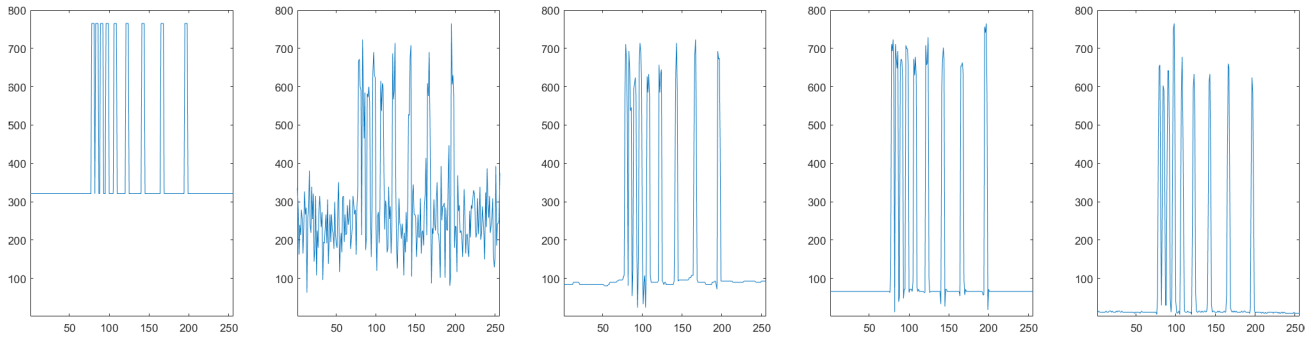


Fig. 8. Sectional curve of vertical simulated point targets. Left to right: original point targets, classical imaging result with speckle noise, PnP-BM3D imaging result, PnP-CNN imaging result, and our proposed conditional diffusion imaging result. We plot the sectional area of vertical simulated point targets as curves.

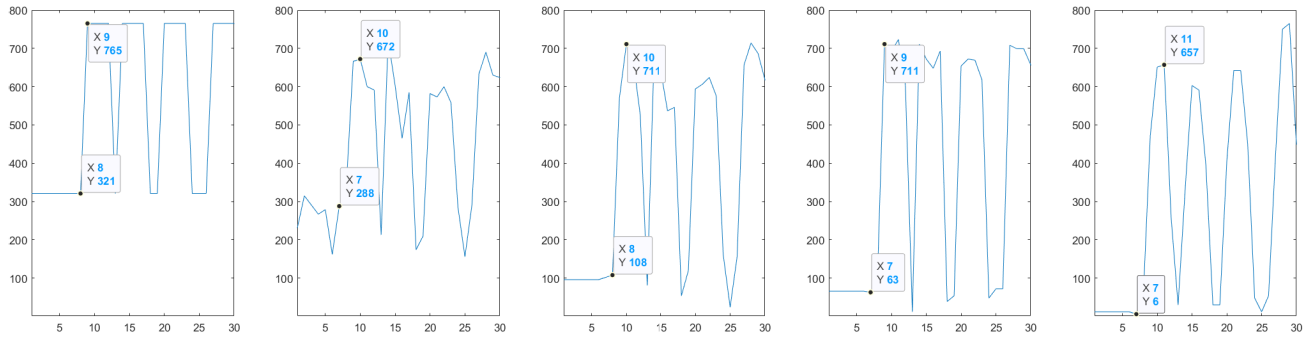


Fig. 9. Sectional curve of vertical simulated point targets, with indicators. Left to right: original point targets, classical imaging result with speckle noise, PnP-BM3D imaging result, PnP-CNN imaging result, and our proposed conditional diffusion imaging result. We plot the sectional area of the first three vertical simulated point targets.

TABLE I

RELATIVE INTENSITY OF POINT TARGET SIGNAL OVER SPECKLED NOISE. WE USE THE INTENSITY OF THE RECOVERED POINT SIGNAL (RATIO OF PEAK TO BOTTOM) **DB**AS THE UNIT OF MEASUREMENT

Algorithms	**horizontal**	**vertical**
FFT-based	8.3965	12.3570
PnP-BM3D	16.0706	16.3689
PnP-CNN	22.3159	21.0506
Proposed	30.5930	40.7086

with a large performance gain, so that in imaging it could give a more distinguishable SAR image of target objects, which is important for downstream remote sensing tasks. Hence, our proposed condition-diffusion-based SAR imaging algorithm performs the best.

B. Simulated Imaging

Here, we adopt the standard test image set for digital image processing as our benchmark and simulate the returned SAR signal profiles according to the SAR imaging physics, as stated in Section II. On the simulated test data, we acquired the corresponding reflected signals from an image. Therefore, with the ground truth as a reference, the normally used image reconstruction quality evaluation indexes, such as PSNR and SSIM, can be implemented.

1) Visually: As shown in Fig. 10, our proposed diffusion model conditional on the generation-based method visually reconstructs the desired image on par with these statistical methods. Compared with statistical methods, the diffusion generative model retrieved more details, such as textures and hairs, while removing a satisfactory amount of speckle noise. Because ground-truth images exist, we can directly sense the reconstruction quality. Although our proposed method performs the best, some fine-grained structures are damaged by simulated speckle noise, which causes inevitable oversmoothness. For better demonstration, we enlarged the local region, as shown in Fig. 11. We can see that the inevitable noise or oversmoothness would be difficult for other algorithms, whereas our proposed method presents a noiseless image with abundant detailed target object features.

2) Numerically: Numerically, our proposed method performs way better than the other two methods with a giant leap on the peak-to-signal-noise ratio (PSNR) index in Table II and Fig. 12. Simultaneously, regarding the structural similarity (SSIM) in Table III and Fig. 13, our proposed algorithm achieves a value comparable to that of PnP methods, but with a few drawbacks. A possible explanation is that the diffusion generative model maintains a potential to create something new, even under strict consistency with the received signal. Combining these two evaluation indices, we can see that the PnP methods seem to be bounded by a Pareto frontier, which restricts the combinatorial optimal of multiple opposed values. There may be a few differences between the denoising ability of BM3D and bias-free CNN, and this difference would affect the reconstruction results of the PnP method. However, with the conditional diffusion generative model, we climb to a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:42:54 UTC from IEEE Xplore. Restrictions apply.

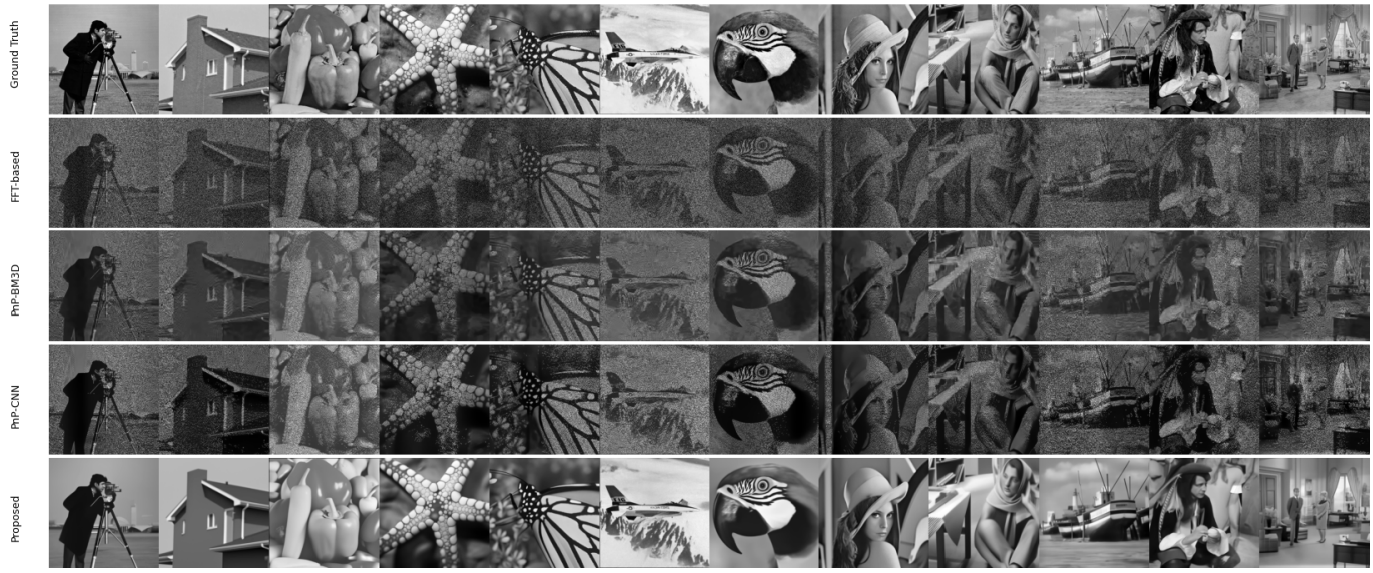


Fig. 10. Simulated results of a standard test image dataset for digital image processing. Above to bottom: the ground-truth image, simulated SAR classical image with speckle noise, the results of two statistical imaging methods PnP framework with the prior model of BM3D and bias-free CNN, and the results of the conditional generative diffusion model. Visually, our proposed conditional diffusion method removed more of these speckle noise than these two statistical methods and reconstructed the most similar intensity with the reference images.

TABLE II

PSNR VALUES OF SET12 PICTURES FOR DIFFERENT ALGORITHMS ON SIMULATED SAR DESPECKLING TASK

Algorithms	01.png	02.png	03.png	04.png	05.png	06.png	07.png	08.png	09.png	10.png	11.png	12.png	Average
FFT-based	16.6856	14.8345	17.2649	15.5571	18.4695	12.7871	16.9616	18.0234	17.9591	15.4529	19.9747	18.1024	16.8394
PnP-BM3D	17.7758	15.7660	18.8409	16.3278	20.3778	13.6141	17.9373	20.2138	19.6519	16.1621	23.4336	19.8594	18.3300
PnP-CNN	17.8328	15.7717	18.8889	16.3634	20.5164	13.5635	17.9602	20.2372	19.6498	16.2015	23.5596	19.9553	18.3750
Proposed	23.8988	22.1671	19.1772	21.4662	22.7811	23.3144	25.4091	22.8330	23.4622	23.6503	22.0140	24.8133	22.9156

TABLE III

SSIM VALUES OF SET12 PICTURES FOR DIFFERENT ALGORITHMS ON SIMULATED SAR DESPECKLING TASK

Algorithms	01.png	02.png	03.png	04.png	05.png	06.png	07.png	08.png	09.png	10.png	11.png	12.png	Average
FFT-based	0.3712	0.2916	0.4014	0.4922	0.5107	0.3136	0.4217	0.4034	0.5007	0.4295	0.4726	0.4708	0.4233
PnP-BM3D	0.7589	0.7786	0.8143	0.7483	0.8602	0.7428	0.7625	0.8218	0.7947	0.6821	0.7642	0.7486	0.7731
PnP-CNN	0.7735	0.7676	0.8177	0.7599	0.8746	0.6210	0.7583	0.8163	0.7930	0.6950	0.7660	0.7730	0.7680
Proposed	0.7851	0.8097	0.7875	0.7394	0.8463	0.7551	0.7695	0.8065	0.7536	0.6898	0.6813	0.7271	0.7626

higher frontier and acquire a better PSNR while maintaining a close SSIM value. Thus, a stronger prior model leverages image quality and semantic details with better ability.

C. Real Signals of Radar-Sat SAR

We used the real signal data of RadarSat-1 SAR as our testbed for the proposed diffusion generative model-based SAR imaging method. RadarSat-1 is a remote sensing satellite developed by the Canadian Space Agency (CSA), which was launched in 1995. RadarSat-1 is a C-band SAR. RadarSat-1 SAR was designed in right-looking and strip-map mode, with a changeable swath ranging from 35 to 500 km. The satellite was placed in a sun-synchronous polar orbit to get global coverage, which emphasizes the observation of the polar regions. The data are partially available in the attachment CD of the book [47]. These imaging regions are all over the city of Vancouver, Canada, and we choose the typical seven scenes among them.

1) Coal and ferry terminals.
2) Vancouver Airport.
3) University of British Columbia and one ship.
4) Stanley Park and city.
5) English Bay ships.
6) Squamish and Garibaldi.
7) Brackendale.

1) Visually: As shown in Fig. 14, the real data would be more complicated than that of the simulated SAR results of pure speckle noise, such as Poisson noise from electromagnetic effects on the receiving sensors and heat-caused noise. As expected, both statistical imaging methods and conditional diffusion methods can eliminate this abnormal noise. Even with the Gaussian noise assumption, these iterative statistical models are designed with a convergence guarantee [6] when confronted with an outdistribution noise. Briefly, the PnP framework progressively divides and conquers the different deviations of noise such that nonuniform noise removal is possible. As for our proposed conditional diffusion method, the generation ability makes it possible to derive the correct feature from noisy pixels and automatically reestablish the contaminated details. For better demonstration, we enlarged the local region, as shown in Fig. 15. We can observe that our

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:42:54 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
ENTROPY VALUES OF DIFFERENT SCENES IMAGING RESULTS FROM RADARSAT SAR SIGNAL DATA FOR DIFFERENT ALGORITHMS

Algorithms	1.terminals	2.airport	3.UBC	4.Stanley	5.Bay	6.Squamish	7.Brackendale	Average
FFT-based	6.5318	6.5840	6.9187	6.5142	6.6649	6.6496	6.8657	6.6756
PnP-BM3D	6.0027	6.3890	6.7070	6.3933	6.4145	6.3553	6.7456	6.4296
PnP-CNN	5.9522	6.6272	6.7051	6.4228	6.3549	6.4933	6.9031	6.4941
Proposed	5.9132	6.6027	6.8430	6.4345	6.3469	6.5169	6.8709	6.5040

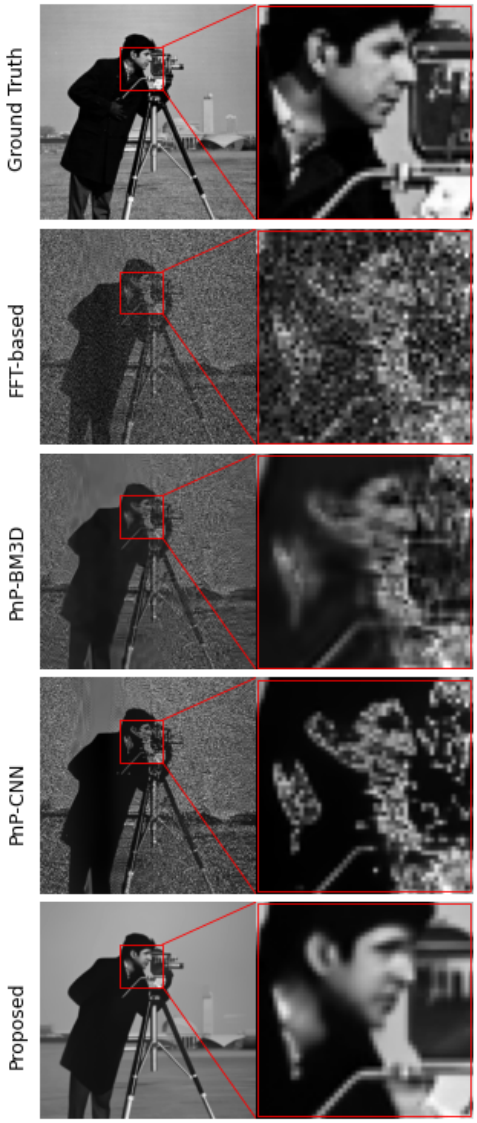


Fig. 11. Simulated results with local details enlarged. Our proposed method presents the best recovery of the image details, like the man's ear, the button on the camera, while these would be noised or oversmothered in other methods.

proposed method can preserve enough target ground features for downstream remote sensing interpretation, while the other two methods, PnP-BM3D and PnP-CNN, are both blurry in the local details of the ground target. For example, in the green box region, there may be buildings or agricultural infrastructure along the river valley. The PnP-BM3D and PnP-CNN algorithms oversmoothed to obtain the desired denoising effect but damaged the target ground features. Our proposed method achieves denoising while preserving the target ground features for remote sensing applications.

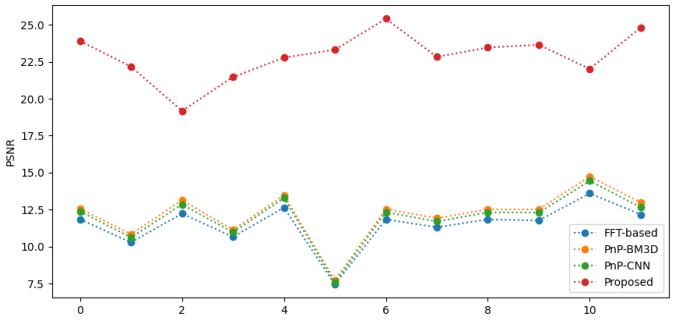


Fig. 12. PSNR values of simulated experiments. X-axis: different pictures in the test image set. Y-axis: PSNR values. We demonstrate four lines of different algorithms, separated by different colors. The green line as our proposed conditional diffusion method gives the highest PSNR values in almost all pictures except for the eleventh picture. These two statistical methods based on the PnP framework give similar PSNR values with the PnP-BM3D method being a little lower.

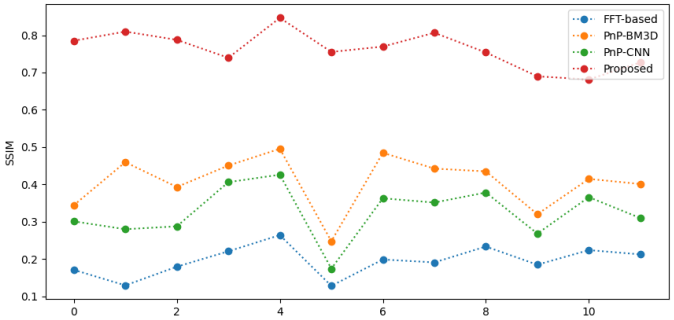


Fig. 13. SSIM values of simulated experiments. X-axis: different pictures in the test image set. Y-axis: SSIM values. We demonstrate four lines of different algorithms, separated by different colors. The green line as our proposed conditional diffusion method would not exceed these two statistical imaging methods that much, but be comparable instead. These three methods perform similarly, so that obey the single variable principle for comparison.

2) Numerically: As the numerical evaluation for the RadarSat SAR imaging experiment, we are only able to use general image quality measurements without reference image [2]. SAR imaging does not have a reference image without speckle noise, because the speckle phenomenon is an intrinsic property. Commonly used image evaluation indices, such as PSNR and SSIM, require the existence of an reference image as the ground truth. Therefore, the evaluation PSNR and SSIM were substituted by the entropy and contrast of the resulting image. To better assess the imagery quality, we need to measure at least two properties of the resulting SAR image: the notability of targets against the background speckle noise and the distinguishability of two near objects.[4] There have been many elaborately designed SAR image assessment

[4]https://www.geospatialworld.net/article/sar-imagery-and-quality-metrics/

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:42:54 UTC from IEEE Xplore. Restrictions apply.

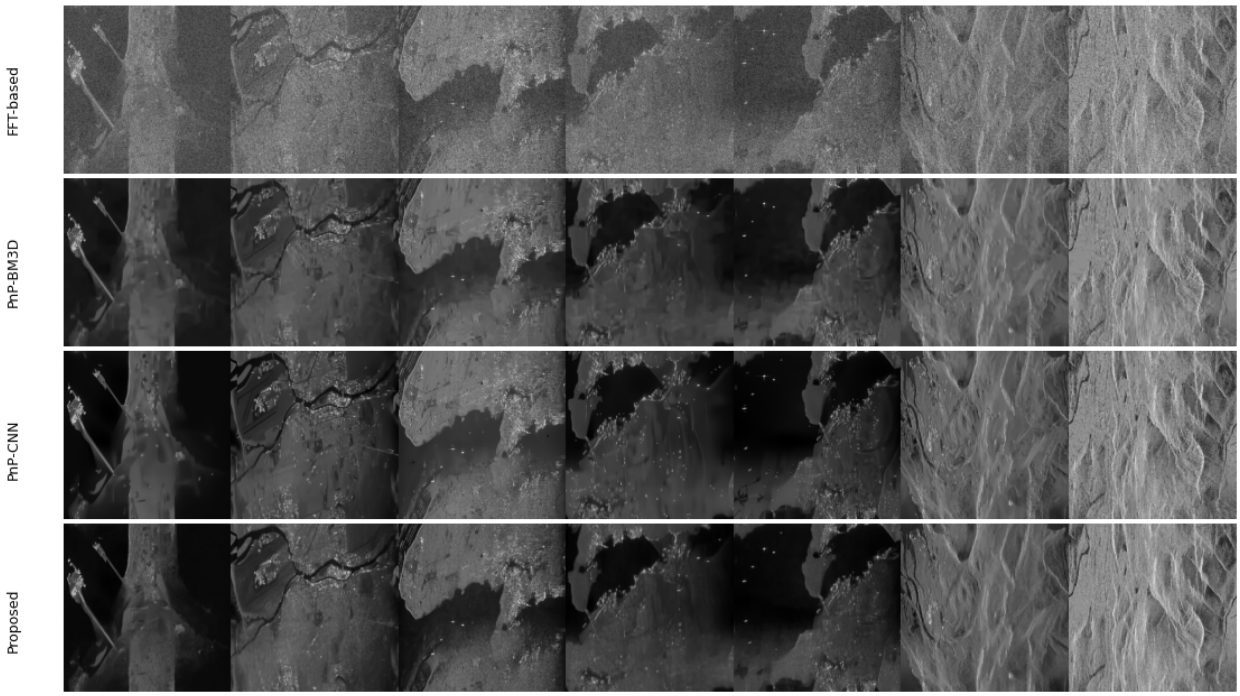


Fig. 14. SAR imaging results of RadarSat-I.

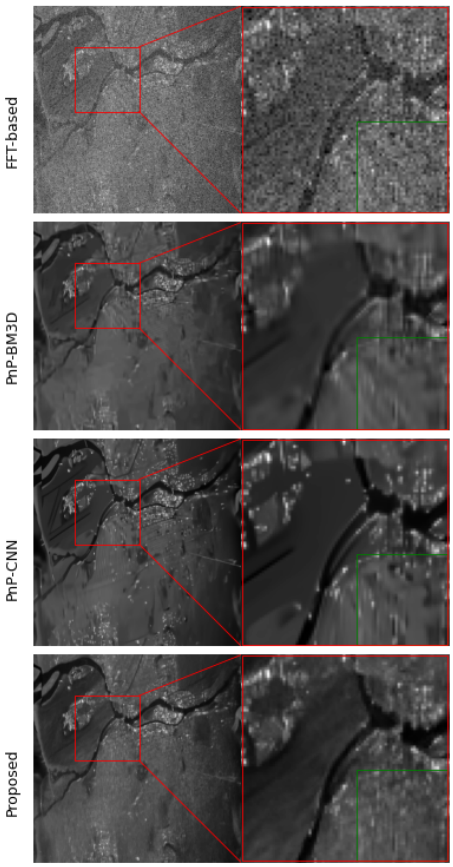


Fig. 15. SAR imaging results of Vancouver Airport, RadarSat-SAR with local details enlarged for visual comparison. The results of other algorithms would be either noised or oversmothered, while our proposed methods would preserve the necessary details while removing the speckle noise, like the green rectangle area.

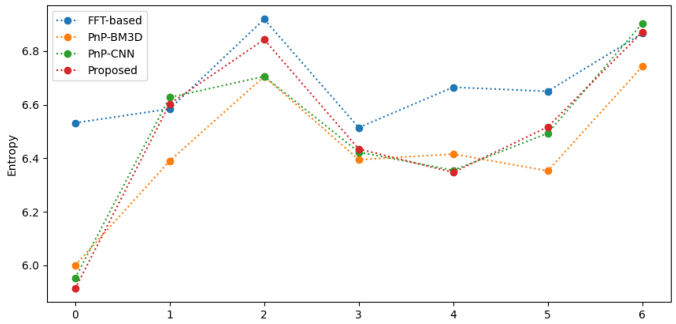


Fig. 16. Entropy values of RadarSat SAR data experiments. X-axis: seven typical scenes of Vancouver City. Y-axis: entropy values. We demonstrate four lines of different algorithms, separated by different colors. The green line as our proposed conditional diffusion method shows a comparable performance compared with these two statistical methods. The average improvement of three advanced methods over The FFT-based signal processing method is not obvious, which possibly can be attributed to the entropy value being dedicated to measuring the extent of chaos only. We should analyze based on both entropy and contrast values.

indexes, such as edge preservation degree based on the ratio of average (EPD-ROA) [48], speckle suppression index (SSI) [49] and speckle suppression and mean preservation index) (SMPI) [50]. Intrinsically, these finely constructed evaluation indexes would relate closely to the entropy and contrast of the resulting image. Entropy is used to measure the intensity of the noise variety. Contrast is used to recognize different regions in the image. By combining these two indexes, we can acquire a general assessment of the SAR image quality for downstream applications [51].

Note that these two evaluation indices are closely related, according to a tradeoff between denoising intensity and object feature preservation. We could easily obtain a very good performance on only one evaluation index while leaving another evaluation index with very bad performance. For example, we could set a stronger denoising ability (the noise variance parameter Gamma) to eliminate more noise, but this

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:42:54 UTC from IEEE Xplore. Restrictions apply.

TABLE V
CONTRAST VALUES OF DIFFERENT SCENES IMAGING RESULTS FROM RADARSAT SAR SIGNAL DATA FOR DIFFERENT ALGORITHMS

Algorithms	1.terminals	2.airport	3.UBC	4.Stanley	5.Bay	6.Squamish	7.Brackendale	Average
FFT-based	0.2040	0.1789	0.2335	0.1834	0.2393	0.1738	0.1716	0.1978
PnP-BM3D	0.1786	0.1316	0.1940	0.1401	0.2028	0.1146	0.1270	0.1555
PnP-CNN	0.1791	0.1336	0.1992	0.1406	0.2032	0.1181	0.1308	0.1578
Proposed	0.7378	0.3122	0.3957	0.4571	0.6392	0.1943	0.1765	0.4161

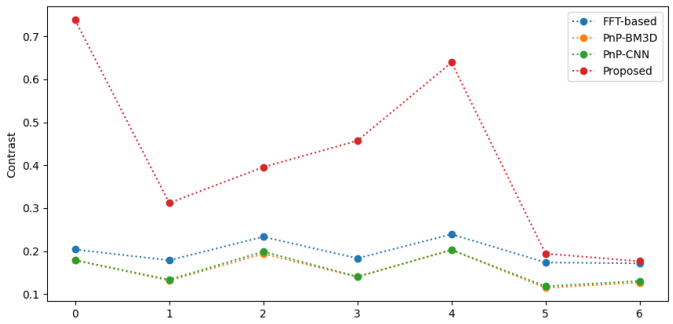


Fig. 17. Contrast values of RadarSat SAR data experiments. X-axis: seven typical scenes of Vancouver City. Y-axis: contrast values. We demonstrate four lines of different algorithms, separated by different colors. The green line as our proposed conditional diffusion method gives the highest contrast values in almost all pictures with a giant margin over others. Note that these two statistical imaging methods would smooth to despeckle so that they were even worse than the FFT-based method.

would harm the possible detailed features that are with similar size to larger noise patches. On the contrary, we could also set a weaker denoising ability to reduce the harm to these useful target details (contrast value), but with a bad denoising performance (entropy value). There is elaborately designed single evaluation index that incorporate the tradeoff between entropy and contrast. However, we prefer to use these intrinsic evaluation index to get more physically clear numerical assess results.

Numerically, while comparable on the entropy metric in Table IV and Fig. 16, our proposed method performs much better than the other two methods on the contrast index in Table V and Fig. 17. This evaluation result indicates that our proposed algorithm achieves a comparable performance on the despeckling task, but with many more details of features preserved or reconstructed. Once again, this demonstrates the powerful generative ability of the diffusion model, which is not contained by these PnP-based methods. Similar to the simulated experiment, the results here reveal that the optimal frontier of our proposed conditional diffusion model is higher than those of PnP-based methods, which makes it possible to better leverage both speckle suppression and edge preservation.

V. CONCLUSION

Given the systematic analysis of the main challenge of the coherent speckle phenomenon during SAR imaging, we propose a novel conditional generative diffusion model method for synthetic aperture radar imaging. Although the diffusion model has no latent space, which hinders its application for conditional signal processing, we technically breakthrough this obturation by fusing the MAP objective function of the SAR inverse imaging problem into the conditional guidance of the diffusion reverse SDE formulation.

Our proposed method aims to handle the training data problem using a transfer learning style scheme. Briefly, we used the learned knowledge from a deep neural network that trained on camera photograph datasets (instead of SAR images). SAR images inherently suffer from speckle noise, whereas natural images such as camera photographs are noiseless and contain sufficient local features to compose the resulting image. The proposed method combines deep learning and statistical methods. The statistical framework is the Bayesian theorem of conditional distributions. The imaging conditions or parameters were used in the likelihood function to approximate the imaging conditions. The deep network performs as a prior function of the image distribution space, which contains all possible image structures. Then, we would find a proximal solution in this prior function space of all possible image spaces to suppress speckle noise that is not included in the image prior space. Therefore, the prior function of the deep network as a deep diffusion generative model is trained on massive image data of natural images, which contains possible bird-eye-view style remote sensing images. We did not use the SAR image dataset for the deep network. This methodology is similar to deep transfer learning, which uses a trained deep network for application to more broader test datasets or different tasks.

We conducted both simulated and real data experiments to test the performance of these SAR imaging methods for the disposal of the speckle noise. We compared the results of the conditional-diffusion imaging method with those of the FFT-based, PnP-BM3D, and PnP-CNN imaging methods. In the simulated experiments, we adopted the PSNR and SSIM for the reference image. For real data experiments, entropy and contrast measurements were performed. The experimental results show that the conditional diffusion SAR imaging method outperforms other SAR imaging methods.

REFERENCES

[1] C. A. Wiley, "Pulsed Doppler radar methods and apparatus," U.S. Patent 3 196 436, Jul. 20, 1965.

[2] A. Moreira, P. Prats-Iraola, M. Younis, G. Krieger, I. Hajnsek, and K. P. Papathanassiou, "A tutorial on synthetic aperture radar," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 1, pp. 6–43, Mar. 2013.

[3] M. Çetin and W. C. Karl, "Feature-enhanced synthetic aperture radar image formation based on nonquadratic regularization," *IEEE Trans. Image Process.*, vol. 10, no. 4, pp. 623–631, Apr. 2001.

[4] C. A. Bouman, *Model-Based Image Processing*. West Lafayette, IN, USA: Purdue Univ., 2015. [Online]. Available: https://engineering.purdue.edu/~bouman/publications/pdf/MBIP-book.pdf

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:42:54 UTC from IEEE Xplore. Restrictions apply.

[5] S. V. Venkatakrishnan, C. A. Bouman, and B. Wohlberg, "Plug-and-play priors for model based reconstruction," in *Proc. IEEE Global Conf. Signal Inf. Process.*, Dec. 2013, pp. 945–948.

[6] S. H. Chan, X. Wang, and O. A. Elgendy, "Plug-and-play ADMM for image restoration: Fixed-point convergence and applications," *IEEE Trans. Comput. Imag.*, vol. 3, no. 1, pp. 84–98, Mar. 2017.

[7] C. J. Pellizzari et al., "Synthetic aperature LADAR: A model-based approach," *IEEE Trans. Comput. Imag.*, vol. 3, no. 4, pp. 901–916, Dec. 2017.

[8] C. J. Pellizzari, T. J. Bate, K. P. Donnelly, G. T. Buzzard, C. A. Bouman, and M. F. Spencer, "Coherent plug-and-play artifact removal: Physics-based deep learning for imaging through aberrations," *Opt. Lasers Eng.*, vol. 164, May 2023, Art. no. 107496.

[9] M. B. Alver, A. Saleem, and M. Çetin, "Plug-and-play synthetic aperture radar image formation using deep priors," *IEEE Trans. Comput. Imag.*, vol. 7, pp. 43–57, 2021.

[10] S. Boyd, "Distributed optimization and statistical learning via the alternating direction method of multipliers," *Found. Trends Mach. Learn.*, vol. 3, no. 1, pp. 1–122, 2010.

[11] D. Zoran and Y. Weiss, "From learning models of natural image patches to whole image restoration," in *Proc. Int. Conf. Comput. Vis.*, Nov. 2011, pp. 479–486.

[12] U. S. Kamilov, H. Mansour, and B. Wohlberg, "A plug-and-play priors approach for solving nonlinear imaging inverse problems," *IEEE Signal Process. Lett.*, vol. 24, no. 12, pp. 1872–1876, Dec. 2017.

[13] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," 2013, *arXiv:1312.6114*.

[14] I. J. Goodfellow et al., "Generative adversarial nets," in *Proc. 27th Int. Conf. Neural Inf. Process. Syst. (NIPS)*, vol. 2. Cambridge, MA, USA: MIT Press, 2014, pp. 2672–2680.

[15] D. J. Rezende and S. Mohamed, "Variational inference with normalizing flows," in *Proc. Int. Conf. Mach. Learn. (ICML)*, vol. 37, 2015, pp. 1530–1538.

[16] J. Guo, B. Lei, C. Ding, and Y. Zhang, "Synthetic aperture radar image synthesis by using generative adversarial nets," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 7, pp. 1111–1115, Jul. 2017.

[17] K. Wang, G. Zhang, Y. Leng, and H. Leung, "Synthetic aperture radar image generation with deep generative models," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 6, pp. 912–916, Jun. 2019.

[18] X. Zhan, X. Zhang, W. Zhang, J. Shi, S. Wei, and T. Zeng, "Near-filed SAR image restoration with deep learning inverse technique: A preliminary study," 2022, *arXiv:2211.14990*.

[19] A. Shastry, A. Smitha, S. George, and P. Jidesh, "Restoration and enhancement of aerial and synthetic aperture radar images using generative deep image prior architecture," *PFG J. Photogramm., Remote Sens. Geoinf. Sci.*, vol. 90, no. 6, pp. 497–529, Dec. 2022.

[20] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Proc. NIPS*, vol. 33. Vancouver, BC, Canada: Curran Associates, 2020, pp. 6840–6851. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2020/file/4c5bcfec8584af0d967f1ab10179ca4b-Paper.pdf

[21] Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole, "Score-based generative modeling through stochastic differential equations," in *Proc. Int. Conf. Learn. Represent.*, 2021. [Online]. Available: https://openreview.net/forum?id=PxTIG12RRHS

[22] M. V. Perera, N. G. Nair, W. G. C. Bandara, and V. M. Patel, "SAR despeckling using a denoising diffusion probabilistic model," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[23] S. Xiao, L. Huang, and S. Zhang, "Unsupervised SAR despeckling based on diffusion model," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2023, pp. 810–813.

[24] X. Bai, X. Pu, and F. Xu, "Conditional diffusion for SAR to optical image translation," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[25] A. Graikos, N. Malkin, N. Jojic, and D. Samaras, "Diffusion models as plug-and-play priors," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 14715–14728. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2022/file/5e6cec2a9520708381fe520246018e8b-Paper.pdf

[26] J. Song et al., "Loss-guided diffusion models for plug-and-play controllable generation," in *Proc. 40th Int. Conf. Mach. Learn.*, vol. 202, pp. 32483–32498. [Online]. Available: https://proceedings.mlr.press/v202/song23k.html

[27] D. Munson, "An introduction to strip-mapping synthetic aperture radar," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process.*, vol. 12, Jun. 1987, pp. 2245–2248.

[28] M. Cheney, "A mathematical tutorial on synthetic aperture radar," *SIAM Rev.*, vol. 43, no. 2, pp. 301–312, Jan. 2001, doi: 10.1137/s0036144500368859.

[29] A. Doerry, "Introduction to synthetic aperture radar," in *Proc. IEEE Radar Conf. (RadarConf)*, Apr. 2019, pp. 1–90.

[30] I. Cumming and J. Bennett, "Digital processing of Seasat SAR data," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, vol. 4, Apr. 1979, pp. 710–718.

[31] S. Chen, M. Xing, T. Yang, and Z. Bao, "A nonlinear chirp scaling algorithm for tandem bistatic SAR," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2013, pp. 2485–2488.

[32] V. K. Madisetti, "A fast spotlight-mode synthetic aperture radar imaging system," *IEEE Trans. Commun.*, vol. 42, nos. 2–4, pp. 873–876, Feb. 1994.

[33] Y. Li et al., "A novel 2-D autofocusing algorithm for real airborne stripmap terahertz synthetic aperture radar imaging," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[34] H.-S. Shin and J.-T. Lim, "Omega-K algorithm for spaceborne spotlight SAR imaging," *IEEE Geosci. Remote Sens. Lett.*, vol. 9, no. 3, pp. 343–347, May 2012.

[35] O. Frey, C. L. Werner, and U. Wegmuller, "GPU-based parallelized time-domain back-projection processing for agile SAR platforms," in *Proc. IEEE Geosci. Remote Sens. Symp.*, Jul. 2014, pp. 1132–1135.

[36] J. W. Goodman, *Speckle Phenomena in Optics: Theory and Applications*. Greenwood Village, CO, USA: Roberts and Company Publishers, 2007.

[37] R. M. Neal, "Annealed importance sampling," *Statist. Comput.*, vol. 11, pp. 125–139, Apr. 2001.

[38] C. Jarzynski, "Equilibrium free-energy differences from nonequilibrium measurements: A master-equation approach," *Phys. Rev. E, Stat. Phys. Plasmas Fluids Relat. Interdiscip. Top.*, vol. 56, no. 5, pp. 5018–5035, Nov. 1997, doi: 10.1103/PhysRevE.56.5018.

[39] J. Sohl-Dickstein, E. Weiss, N. Maheswaranathan, and S. Ganguli, "Deep unsupervised learning using nonequilibrium thermodynamics," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 2256–2265. [Online]. Available: https://proceedings.mlr.press/v37/sohl-dickstein15.html

[40] Y. Song and S. Ermon, "Generative modeling by estimating gradients of the data distribution," in *Proc. Adv. Neural Inf. Process. Syst.*, H. Wallach, H. Larochelle, A. Beygelzimer, F. d'Alché-Buc, E. Fox, and R. Garnett, Eds., vol. 32. Red Hook, NY, USA: Curran Associates, 2019, pp. 11918–11930. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2019/file/3001ef257407d5a371a96dcd947c7d93-Paper.pdf

[41] P. Dhariwal and A. Q. Nichol, "Diffusion models beat GANs on image synthesis," in *Proc. Adv. Neural Inf. Process. Syst.*, A. Beygelzimer, Y. Dauphin, P. Liang, and J. W. Vaughan, Eds., 2021, pp. 8780–8794. [Online]. Available: https://openreview.net/forum?id=AAWuCvzaVt

[42] H. Go et al., "Towards practical plug-and-play diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 1962–1971.

[43] B. Kawar, M. Elad, S. Ermon, and J. Song, "Denoising diffusion restoration models," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 23593–23606.

[44] D. Kostadin, F. Alessandro, K. Vladimir, and E. Karen, "Image denoising by sparse 3-D transform-domain collaborative filtering," *IEEE Trans. Image Process.*, vol. 16, no. 8, pp. 2080–2095, Aug. 2007.

[45] S. Mohan, Z. Kadkhodaie, E. P. Simoncelli, and C. Fernandez-Granda, "Robust and interpretable blind image denoising via bias-free convolutional neural networks," in *Proc. Int. Conf. Learn. Represent.*, 2020. [Online]. Available: https://openreview.net/forum?id=HJlSmC4FPS

[46] P. Chatterjee and P. Milanfar, "Is denoising dead?" *IEEE Trans. Image Process.*, vol. 19, no. 4, pp. 895–911, Apr. 2010.

[47] I. G. Cumming and F. H.-C. Wong, *Digital Processing of Synthetic Aperture Radar Data: Algorithms and Implementation*. Norwood, MA, USA: Artech House, 2005.

[48] H. Feng, B. Hou, and M. Gong, "SAR image despeckling based on local homogeneous-region segmentation by using pixel-relativity measurement," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 7, pp. 2724–2737, Jul. 2011.

[49] Y. Sheng and Z.-G. Xia, "A comprehensive evaluation of filters for radar speckle suppression," in *Proc. Int. Geosci. Remote Sens. Symp.*, vol. 3, Jun. 1996, pp. 1559–1561.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:42:54 UTC from IEEE Xplore. Restrictions apply.

[50] A. Shamsoddini and J. Trinder, "Image texture preservation in speckle noise suppression," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci. ISPRS Arch.*, vol. 38, pp. 239–244, Jan. 2010.

[51] Q. Mao, Y. Li, and Y. Zhu, "A hierarchical feature fusion and attention network for automatic ship detection from SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 13981–13994, 2024.



Zhongqi Wang (Graduate Student Member, IEEE) received the B.S. degree from the Department of Automation, Tsinghua University, Beijing, China, in 2020. He is currently pursuing the Ph.D. degree in signal and information processing with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing.

His research interests include synthetic aperture radar (SAR) imaging, computational imaging, and deep learning.



Chong Song (Member, IEEE) received the B.S. degree from Zhengzhou University, Zhengzhou, Henan, China, in 2018, and the Ph.D. degree in signal and information processing from the University of Chinese Academy of Sciences, Beijing, China, in 2023.

He is currently an Assistant Research Professor with the National Key Laboratory of Microwave Imaging Technology, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing. His research interests include multichannel synthetic aperture radar (SAR) signal processing and moving target indication.



Zekun Jiao received the B.S. degree in electronic engineering from the University of Science and Technology of China, Hefei, China, in 2014, and the Ph.D. degree in signal and information processing from the University of Chinese Academy of Sciences, Beijing, China, in 2019.

He is currently with the National Key Laboratory of Microwave Imaging Technology, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing. His research interests include synthetic aperture radar (SAR) tomography and holography.



Bingnan Wang (Senior Member, IEEE) received the Ph.D. degree in signal and information processing from the University of Chinese Academy of Sciences, Beijing, China, in 2011.

He is currently a Research Professor with the National Key Laboratory of Microwave Imaging Technology, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing. His research interests are focused on signal processing of advanced synthetic aperture radar (SAR) systems, including the theory and applications of SAR interferometry, digital array SAR signal processing, and unmanned aerial vehicle (UAV) SAR imaging and detection.



Maosheng Xiang received the M.S. degree in aerospace electronic engineering from Northwestern Polytechnical University, Xi'an, China, in 1989, and the Ph.D. degree from the Institute of Remote Sensing Applications, Chinese Academy of Sciences, Beijing, China, in 1999.

He is currently a Research Professor with the National Key Laboratory of Microwave Imaging Technology, Aerospace Information Research Institute, Chinese Academy of Sciences. His research interests are microwave remote sensing signal processing and applications and interferometric synthetic aperture radar (SAR) system technology.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:42:54 UTC from IEEE Xplore. Restrictions apply.